SRNDNA Ultimatum: current results

Draft scientific review | 2 November 2026 | Repository: DVS-Lab/srndna-ultimatum

Complete: 24 PALM jobs, 96 signed result summaries, 47 participants (25 younger, 22 older), 10,000 sign-flips per test. No failures, exclusions or L1/L2 refits. All tracked PALM inventory hashes were verified before rendering.

Main message: choice depends strongly on offer amount; the clearest current imaging results concern full-sample task responses and an age difference in the human-computer response contrast. The original DMN age narrative is not stable at the paper-matched cluster threshold.

Result	Current evidence	Figure status
Social-choice offer effect	beta 1.801; SE .237; p < .001	Main behavioral figure
Age x offer x similarity	p = .230; no detected moderation	Report alongside behavior
Positive full-sample offer slope	PALM Z3.1 p = .0052; 45 voxels	Foundational task effect
Human > computer task response	PALM Z3.1 p = .0002; 270 voxels	Foundational partner effect
Computer > human task response	PALM Z3.1 p = .0012; 155 voxels	Separate regional effect
Younger > older: human-computer	PALM Z3.1 p = .0163; 25 voxels	Current age candidate
Younger > older: DMN full-RT	Z3.1 p = .2526; Z2.6 p = .0468	Threshold-sensitive supplement
Younger > older: task-wide offer	Z3.1 p = .1017; Z2.6 p = .0352	Threshold-sensitive supplement
Partner ratings	Human/computer differences; no reliable age effect	Behavioral context / supplement
Gu-style individual parameters	Recovery gates not cleared	No imaging covariates released

Spatial FWE is per signed contrast, not across the full model search. PALM has separate age variance groups and does not reproduce FLAME VARCOPE weighting. Offer amount confounds fairness and own monetary payoff.

Interpretation and remaining work

Paper consistency: retain categorical younger/older groups and the four reported controls: sex, tSNR, mean framewise displacement and corrected mean task RT. The new PALM batch uses verified motion/tSNR values. Full-sample means use an all-ones column with centered age and nuisance columns. FLAME is the submitted estimator; PALM is an unequal-variance robustness analysis, not an automatically adopted replacement.

Historical comparison: at threshold 3.1, PALM significant regions for positive offer slope, human > computer, computer > human and the human-computer age contrast are completely contained in significant FLAME and randomise regions. Historical models use some different covariates/masks; this is spatial convergence, not a fully matched comparison of estimators.

Timing-dependent network means: pooled-post DMN similar > dissimilar offer coupling: p = .0424, 10 voxels. ECN dissimilar > similar: pooled-post p = .0298, 10 voxels; partner-post p = .0121, 13 voxels (8 overlap). These are full-sample effects, not restored age findings. Precise anatomical labels require atlas review.

DMN sensitivity: the lower-threshold full-RT age region contains the same 49 voxels as the earlier coverage-corrected extraction. The descriptive age gap increases from 25.2 without group motion to 36.9 with group motion; selected-region bars are not independent inference.

Other work preserved: the parietal brain-behavior interaction survived earlier robust/deletion diagnostics, but is a different estimand from a direct age difference. Its coverage-corrected batch has not been pushed. The original broader PALM collection remains at its last pushed snapshot. vmPFC ROI results are deliberately omitted from this paper figure set.

Computational behavior: empirical bias-model fits still have divergences; six of eight conditional recovery fits need diagnostic review. Norm sensitivity, learning rate, initial norm and choice consistency are not currently cleared as individual-difference imaging covariates.

Before submission: settle the primary timing specification; finish matched-design comparisons if reporting estimator robustness; inspect anatomical/influence diagnostics for new primary maps; finalize figure extraction/captions; revise the manuscript and response. The manuscript has not yet been updated. OpenNeuro ds003745 version 2.2.0 is published.

What was repaired and what remains exploratory

Event provenance: the sub-144 Ultimatum/Shared Reward event-file problem was repaired from the retained source records. Sub-143 did not require synthetic event_RT duplicate rows to recover its behavioral RT values. The dataset and the paper analysis sample are distinct: 50 recruited, 47 retained for this paper.

Dedicated RT regressors: the historical RT EVs omitted block-first responses. The corrected full-display model includes all responded trials in its dedicated constant and RT-height terms. Full-run activation and network models were regenerated. Fixed display, decision plus pooled post-response, and decision plus partner post-response specifications remain separate; they are not interchangeable estimates of decision time.

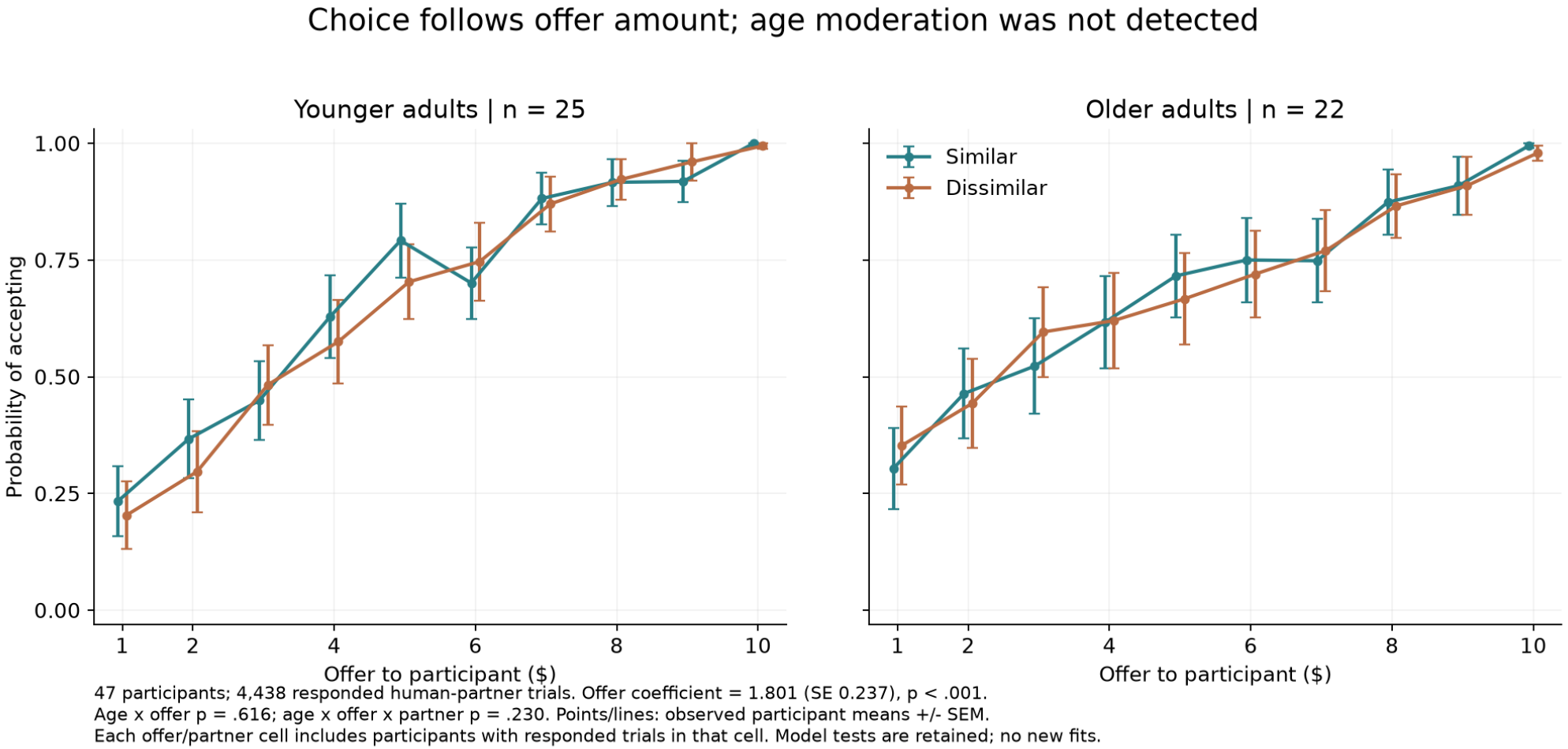
Coverage and group inputs: the strict whole-brain mask requires supported finite estimates and positive VARCOPEs in all 94 L1 runs and all 47 L2 estimates. It contains 59,297 native voxels, versus 64,444 in the historical mask. All participants are retained. Motion/tSNR columns were reconciled against independently verified measurements; corrected mean RT was retained. Continuous signed network maps were preserved.

Ratings and choices: 42 participants have pre-task ratings, 42 post-task ratings and 41 paired records, using the last complete attempt. Humans were rated fairer and more likeable than the computer. Similar-minus-dissimilar choice-slope associations with pre-task likeability (Spearman rho = -.405) and post-task anger (rho = .397) survive the retained BH adjustment (q = .0366 each), but corresponding Pearson associations do not. Trial-level rating moderation is nonsignificant for pre-task fairness (p = .868) and likeability (p = .834). These are exploratory associations, not an age mechanism.

Norm-model diagnostics: the retained stable bias extension has 2 divergences in the full fit and 7 in the run-1 fit. Improved convergence does not establish parameter recovery; the conditional screen still flags six of eight recovery fits. Individual parameters are not currently used to support imaging claims.

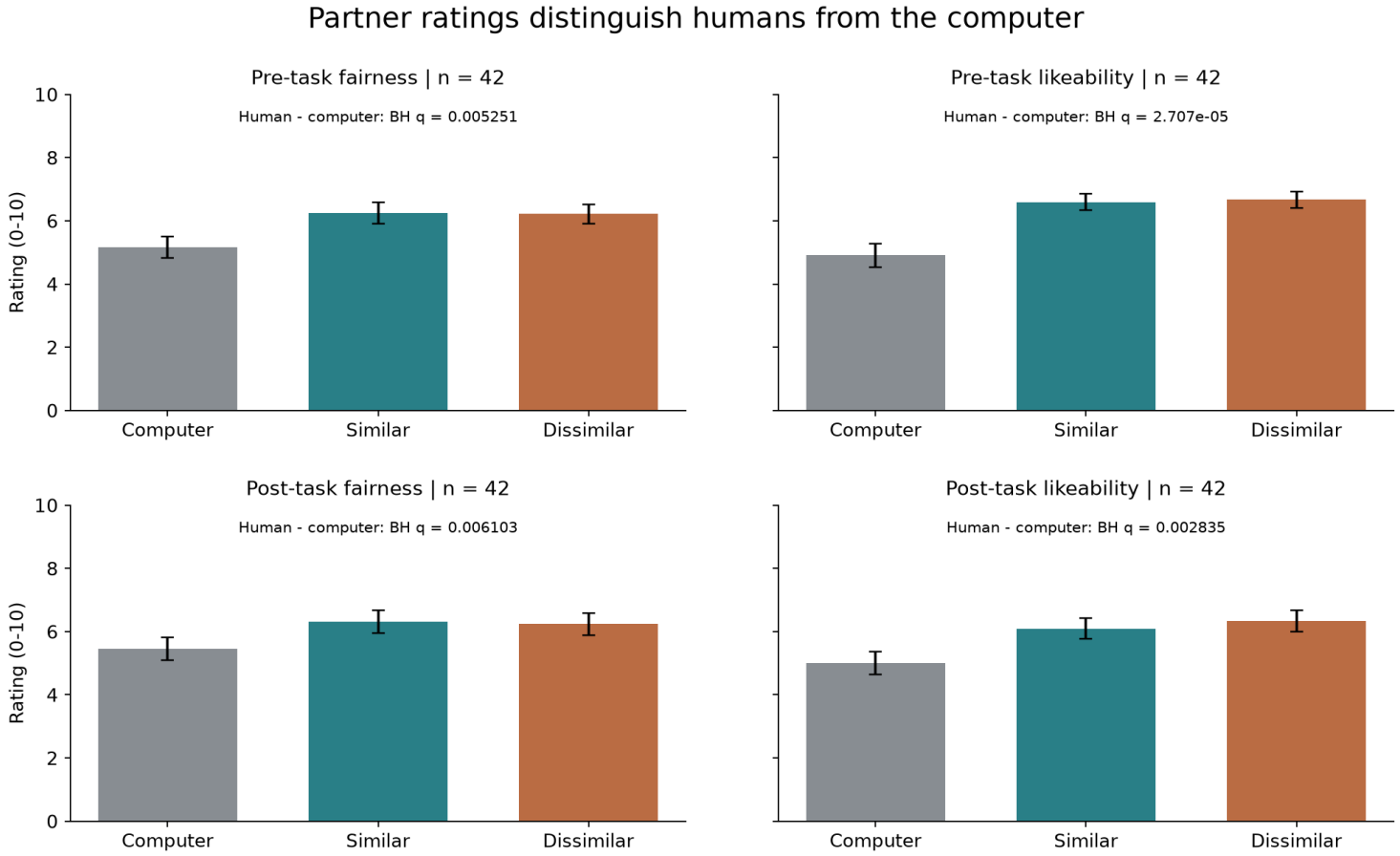
Claims to avoid: no proof of behavioral equivalence; no isolated neural fairness signal distinct from own payoff; no restored original ECN age effect; no robust DMN age claim based on selecting Z = 2.6; no causal or directional interpretation of PPI; and no relabeling older selected-region bars as estimates from the new 25-voxel age region.

Draft Figure 1 | Corrected social-choice behavior



Acceptance increased with offer size. The corrected random-intercept/random-offer-slope model converged without singularity. No reliable age or age-by-partner moderation was detected; this does not establish equivalence. Points show observed participant means and SEM, rather than a conditional logistic curve at zero random effects. Similar/dissimilar refer to proposer age relative to the participant. Offer amount also changes the participant's monetary payoff.

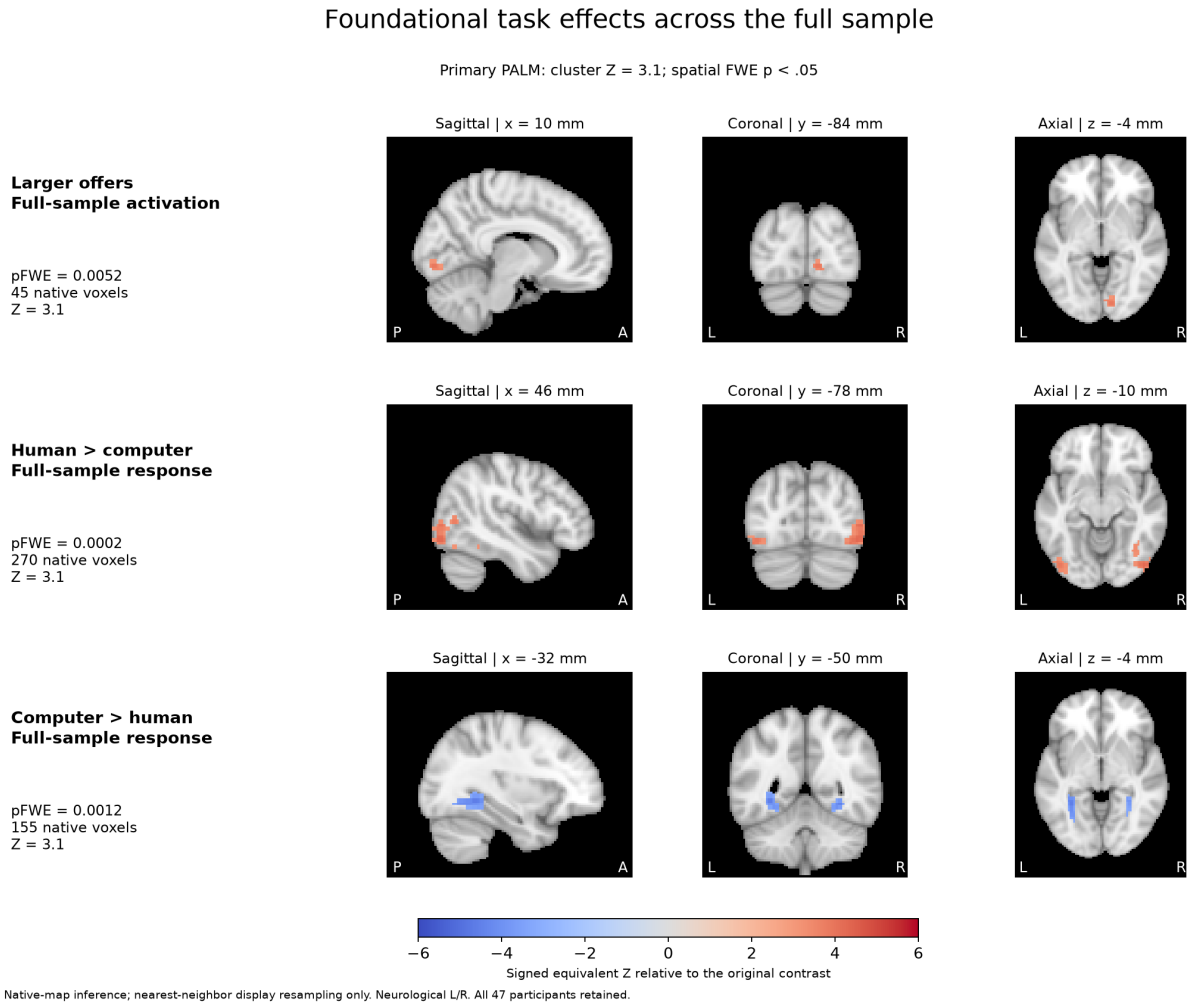
Draft Figure 2 | Partner ratings



Means +/- participant SEM. Last complete attempt retained. Pre/post samples overlap (41 paired participants).
Similar versus dissimilar fairness/likeability contrasts do not survive BH adjustment. These plots do not test changes caused by the task.

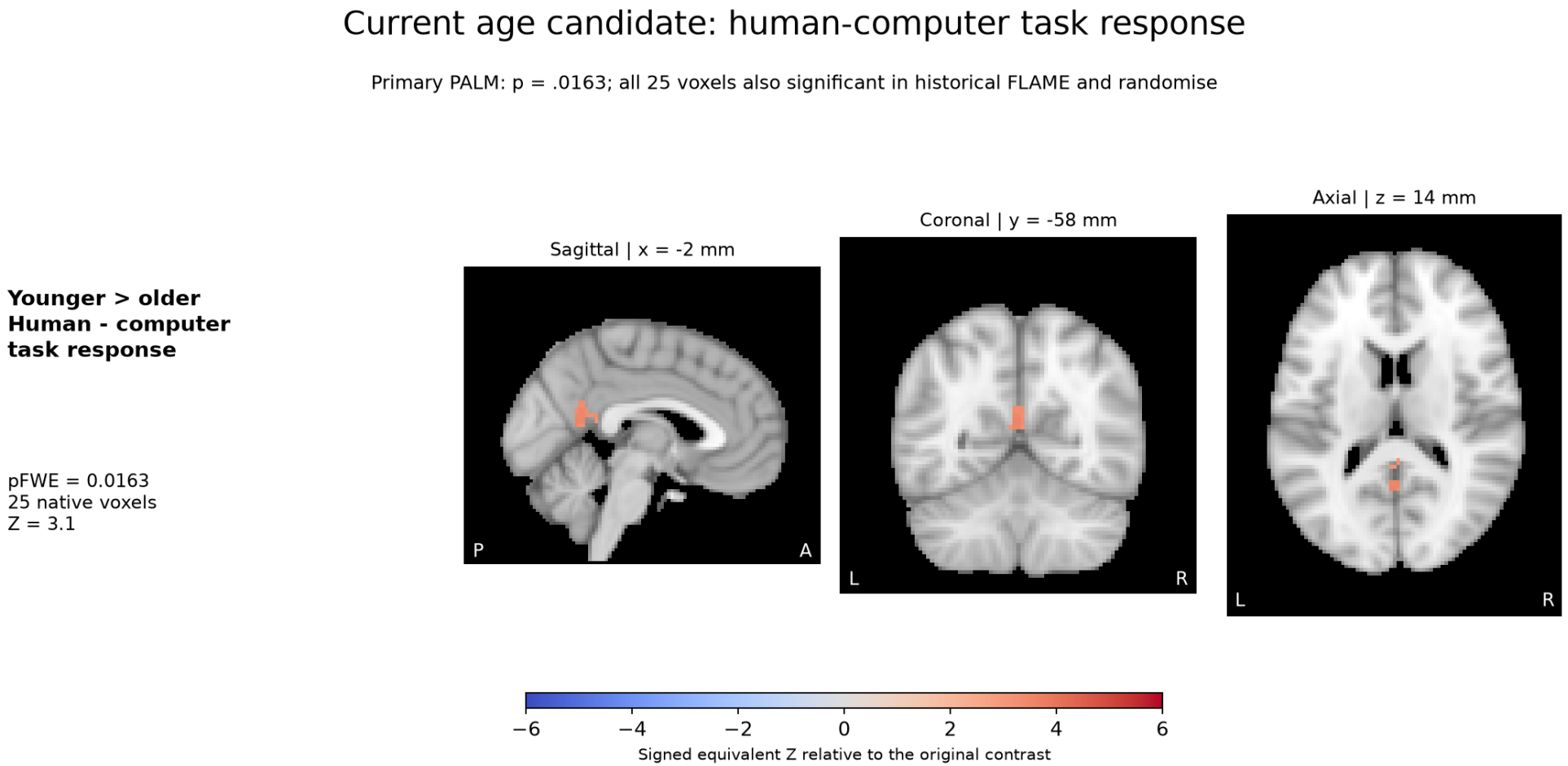
Humans were rated fairer and more likeable than the computer before and after the task. The corresponding similar-minus-dissimilar comparisons are nonsignificant after BH adjustment. These are ratings in a subsample, not evidence that partner ratings mediate an age effect.

Foundational task effects across the full sample



All three effects survive PALM and overlap the historical FLAME/randomise significant regions. Positive offer-related activity is largely posterior, not an established vmPFC/ventral-striatal fairness effect. Human/computer response effects concern the full task display and do not isolate offer sensitivity.

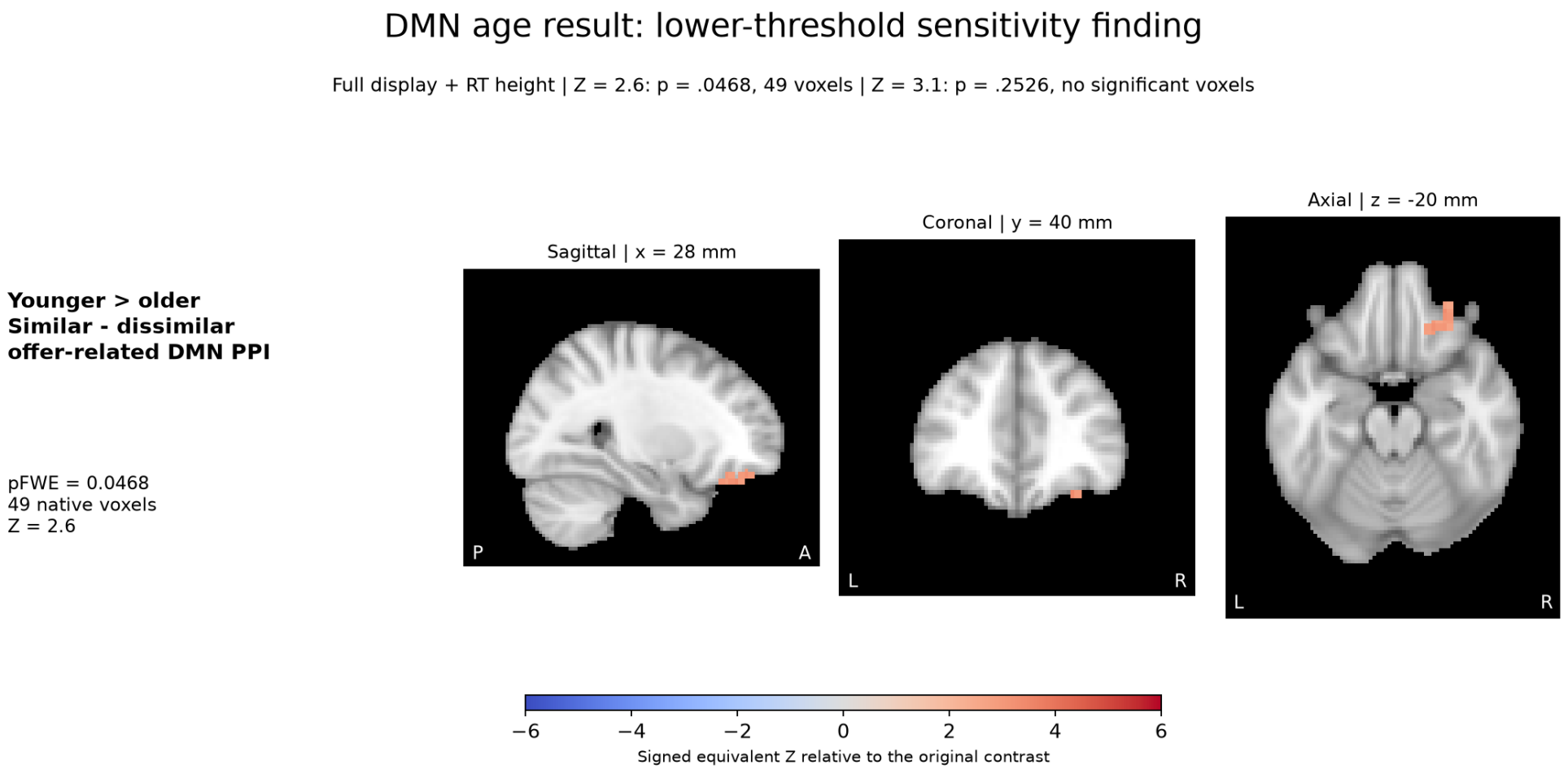
Current age candidate: human-computer task response



Native-map inference; nearest-neighbor display resampling only. Neurological L/R. All 47 participants retained.

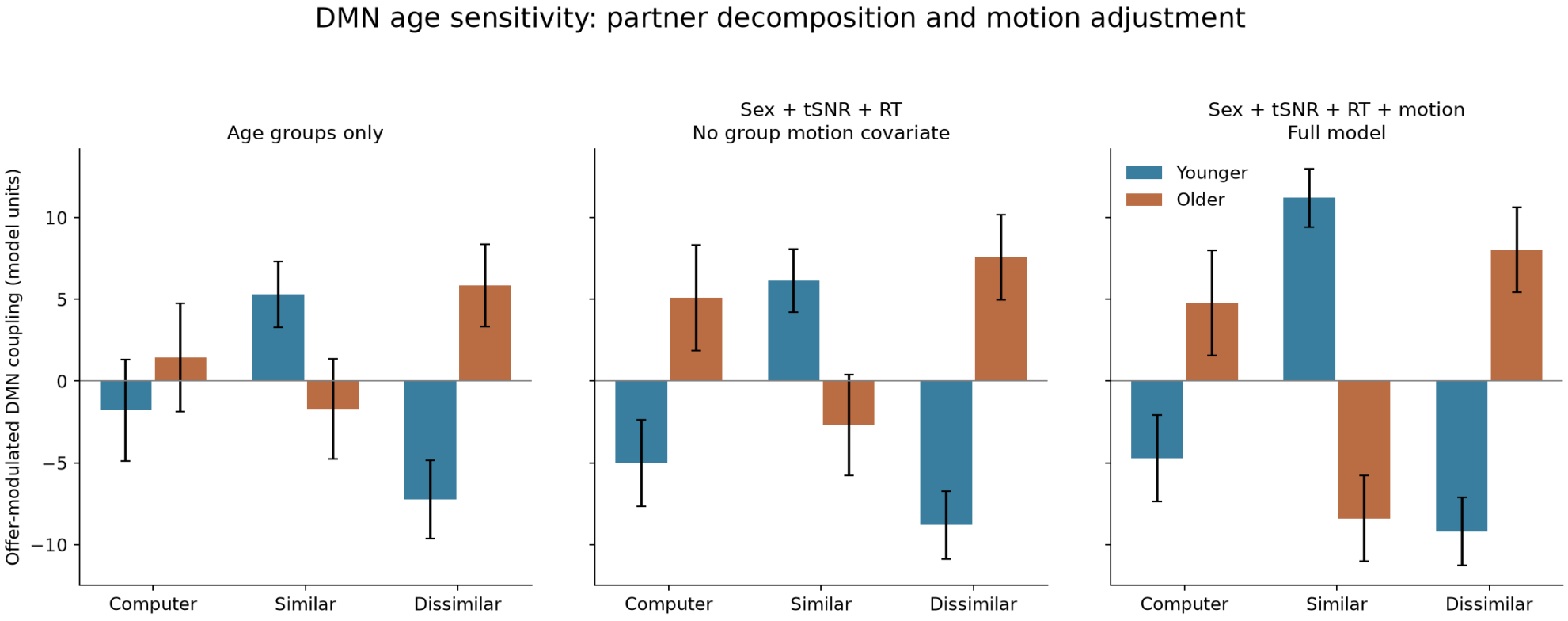
A posterior midline age difference in the human-minus-computer task-response contrast. This is not a fairness-slope age effect. Historical methods use some different covariates/masks. Participant-level bars for this exact new 25-voxel region require a fresh Linux extraction; older-region bars are not relabeled here.

DMN age result: lower-threshold sensitivity finding



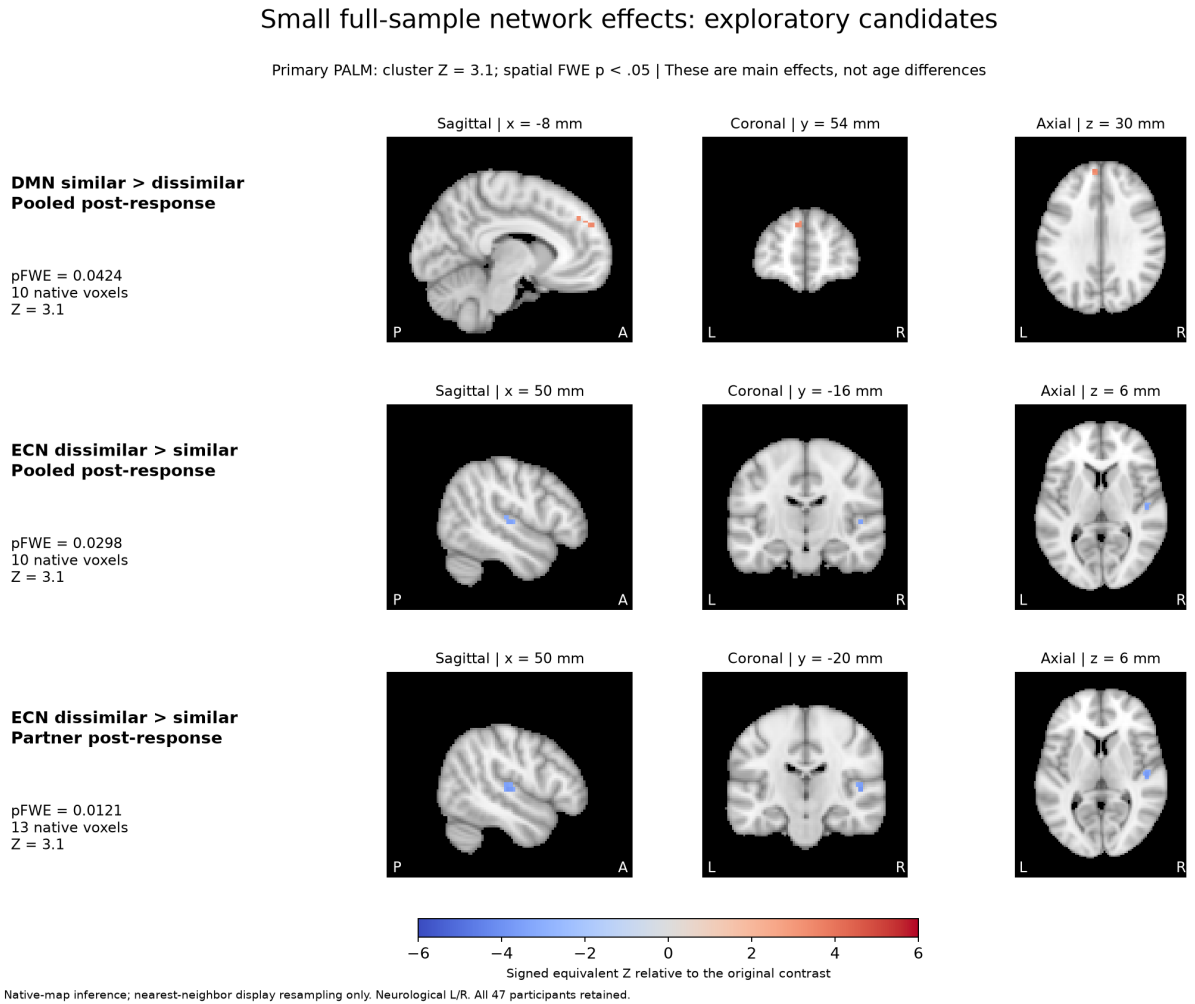
The 49-voxel map exactly matches the previous coverage-corrected PALM region. The result does not survive the paper-matched Z = 3.1 threshold under any timing specification and should not be described as a robust primary age effect.

Sensitivity Figure | DMN three-partner bars



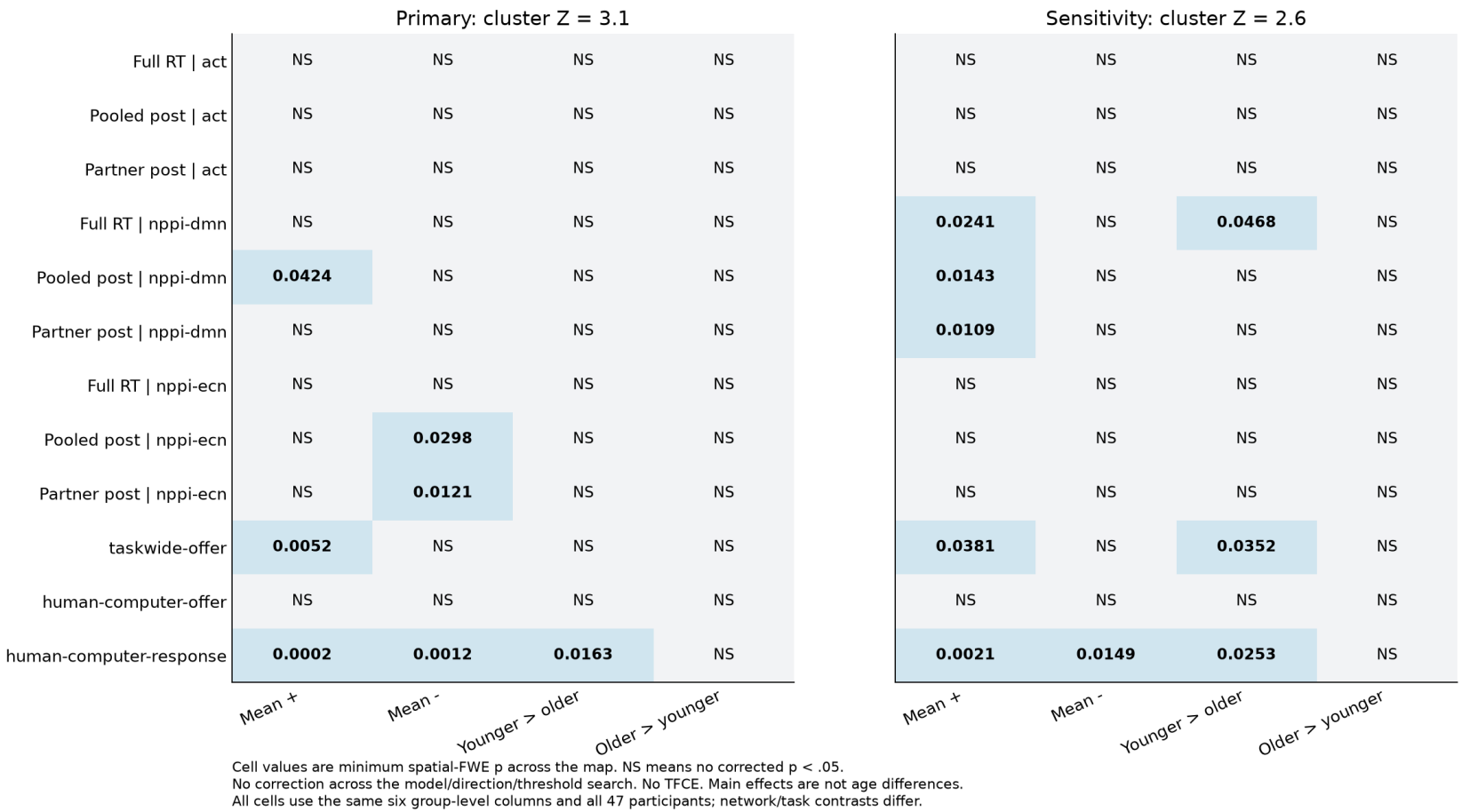
The age-gap direction is present before group motion adjustment, but its magnitude increases substantially after adjustment. The apparent crossover corresponds to positive offer-related coupling for younger proposers and negative coupling for older proposers in both participant groups. The computer condition supplies context and is not part of the similar-minus-dissimilar contrast. No new ROI significance test is performed.

Small full-sample network effects: exploratory candidates



Offer-modulated partner differences in network coupling. The two ECN maps overlap in 8 voxels. These effects are absent at Z = 3.1 in the full-RT model. Precise tissue/atlas localization and influence checks remain before interpretation.

Audit Figure | Complete PALM inventory



All planned results are shown, including null findings. The cluster-Z = 3.1 human-computer task-response age contrast is the only primary PALM age finding. The DMN age effect and task-wide offer age effect occur only at the lower cluster threshold. Full-sample network effects depend on timing specification.